

Assignment 4

due: 2026-10-07

Problem 1: Let $n \in \mathbb{N}$ and let $D = \langle \sigma, \tau \rangle$ be the dihedral group of order $2n$ as in Example 2.6.6 in the notes. (So the order of σ is n .)

- Suppose that $n = p$ is an odd prime number. Find a 2-Sylow subgroup P of D .
- Suppose that $n = 2p$ where p is an odd prime number. Find a 2-Sylow subgroup P of D in this case.
- In the two cases above, decide whether or not P is normal in D .

Problem 2:

Let G be a group and let $H, K \subseteq G$ be subgroups of G . Suppose that H and K are both normal in G , and that $H \cap K = \{1\}$. Recall that HK is a subgroup of G . Prove that the natural map $H \times K \rightarrow HK$ given by $(h, k) \mapsto h \cdot k$ is a group isomorphism.

Recall that the group structure on the cartesian product is given for $(h, k), (h', k') \in H \times K$ by:

$$(h, k) \cdot (h', k') = (hh', kk'), \quad (h, k)^{-1} = (h^{-1}, k^{-1}) \quad \text{and} \quad 1_{H \times K} = (1_H, 1_K);$$

this is the **direct product** of H and K .

Problem 3:

Let $\varphi : G \rightarrow H$ be a group homomorphism, and suppose that $N \subseteq H$ is a normal subgroup of H . Prove that $\varphi^{-1}(N) = \{g \in G \mid \varphi(g) \in N\}$ is a normal subgroup of G .

Problem 4:

Suppose that G and G' are groups, let H, K be subgroups of G , and let H', K' be subgroups of G' . Assume that

- H normalizes K and H' normalizes K' ;
- $G = \langle K, H \rangle = KH$ and $G' = \langle K', H' \rangle = K'H'$;
- $K \cap H = \{1\}$ and $K' \cap H' = \{1\}$;
- there are group isomorphisms $\varphi : H \xrightarrow{\sim} H'$ and $\psi : K \xrightarrow{\sim} K'$ such that for all $h \in H$ and $k \in K$,

$$\psi(\text{Inn}_h(k)) = \text{Inn}_{\varphi(h)}(\psi(k)).$$

(Since H normalizes K , the restriction of Inn_h to K is an automorphism of K for each $h \in H$; similarly $\text{Inn}_{h'}$ restricts to an automorphism of K' for each $h' \in H'$. The last hypothesis says that φ and ψ intertwine these two families of automorphisms.)

Prove that there is a group isomorphism $\Phi : G \rightarrow G'$ given for $k \in K, h \in H$ by

$$\Phi(kh) = \psi(k)\varphi(h) \in K'H' = G'.$$

Problem 5:

For a prime number p , write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ and let

$$H_p = \left\{ \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \mid a, b, c \in \mathbb{F}_p \right\}$$

so that H_p is a subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$ of order p^3 . (You should at least think through why this is so, though you needn't submit the details).

a. Prove that H_2 is isomorphic to $D_8 = D_{2 \cdot 4}$, the dihedral group with 8 elements.

Hint: find $\sigma, \tau \in H_2$ with $o(\sigma) = 4$, $o(\tau) = 2$ which have the property that $\tau\sigma\tau = \sigma^{-1}$. Then $H_2 = \langle \sigma \rangle \cdot \langle \tau \rangle$. Now use the solution to an earlier problem.

b. Show that H_p is a p -Sylow subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$.

Problem 6:

Let G be a finite group, let p be a prime number, and let $P \in \mathrm{Syl}_p(G)$. Let

$$H = N_G(P) = \{g \in G \mid \mathrm{Inn}_g P = P\}$$

be the normalizer of P in G . Prove that $N_G(H) = H$. (In words: the normalizer of a Sylow p -subgroup is self-normalizing).

Problem 7: Prove that there is no simple group of order 56.

Problem 8: Let H be a normal subgroup of G and let $P \in \mathrm{Syl}_p(H)$. Prove that $G = H \cdot N_G(P)$.

Hint: For $g \in G$, show that gPg^{-1} is again a Sylow p -subgroup of H . Then use conjugacy of Sylow p -subgroups *within* H to find $h \in H$ with $gPg^{-1} = hPh^{-1}$, and conclude that $h^{-1}g \in N_G(P)$.